Chapter 23

Applications Of Inversive Geometry

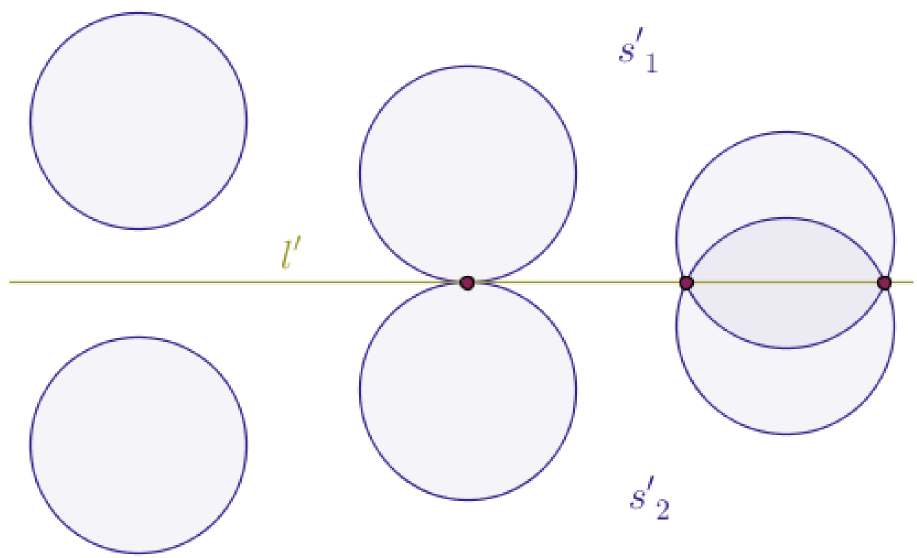
Two For Tea And Tea For Two

In Chapter 17, Inversive Geometry and Verbal College Entrance Exams, we looked at the reverse-order or the reverse-time line problem-solving approach in great detail. In order to demonstrate the utility of such an approach on yet another short and sweet problem, let us consider the following scenario.

Problem 29. Given two, distinct and fixed, coplanar circles, $s_{1,2}$, of distinct radii, $r_1 \neq r_2$, not located one inside of the other, find a way to transform these circles into two circles, $s'_{1,2}$, of equal radii such that $r'_1 = r'_2$.

In this problem, as it is the case with all other problems in this book, we are only allowed to use the Euclidean tools, a compass and a straight edge.

Solution. As the reverse-order problem-solving approach prescribes, we assume, for a moment, that the problem at hand has been solved already. Exactly how this problem has been solved is of no concern to us at this stage (Figure 23.1):



What type of tactically useful information can be extracted from or synthesized based on the configuration of the given objects sought-after?

More formally, which particular IF P THEN Q implication can help us unwind the above solved state of the problem into the original statement?

Well, the following implication seems to be a top contender: if two circles $s'_{1,2}$ have equal radii $r'_1 = r'_2$ then these circles are the classical straight-ling reflections of each other in a straight line, l', whose location is uniquely determined by the location of the centers $S'_{1,2}$ of the said circles. The role of such a straight line, l', is played by the perpendicular bisector of the line segment $S'_1S'_2$.

But we remember that topologically circles and straight lines are equivalent in the following sense:

> *inversion with respect to a circle transforms straight lines that do not pass through the center of inversion into circles that do pass through the center of inversion and it transforms circles that pass through the center of inversion into straight lines that do not pass through the center of inversion*

Thus, still working in reverse order, we realize that our straight line l' is none other than an image of a circle, l, that passes through a point, P, that plays the role of the center of an or some inversion $\mathcal{I}$ with positive (!) power in a circle, c, whose center is located exactly at the point P.

Moreover, our circles $s'_{1,2}$ should be the images of the given circles $s_{1,2}$ under that inversion $\mathcal{I}$.

A Notational Agreement

Let us agree to symbolically record an inversion of a plane with positive power centered at a point O with a radius r as follows:

$$\mathcal{I}^{+}_{O,r} \tag{1}$$

and let us agree to symbolically record an inversion of a plane with negative power centered at a point O with a radius r as follows:

$$\mathcal{I}^{-}_{O,r} \tag{2}$$

Then, the fact that such an inversion transforms an object s into its image s' can be recorded as usual:

$$\mathcal{I}^{+}_{O,r}(s) = s'$$

or:

$$\mathcal{I}^{-}_{O,r}(s) = s'$$

In such a notation the ambient plane is assumed and it is taken to be anonymous. If there is a need to differentiate between various planes then the name of the plane can be moved into the superscript as follows:

$$\mathcal{I}^{\pi,+}_{O,r}(s) = s'$$

which tells us that the plane named $\pi$is inverted with positive power with respect to a circle that has the point O as its center and r as its radius, etc.

Using the notation specified above, we see that we so far demand that for some inversion $\mathcal{I}$ it should be the case that:

$$\mathcal{I}^{+}_{P,r}(s_1) = s'_1$$

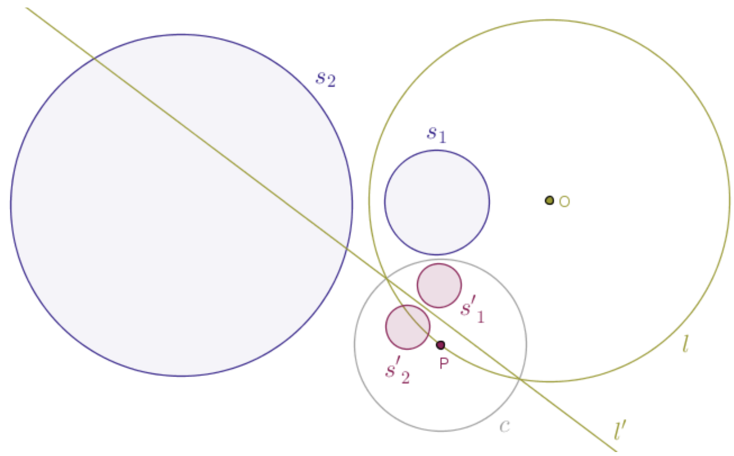
and that:

$$\mathcal{I}^{+}_{P,r}(s_2) = s'_2$$

But if the circles ${s'}_{1,2}$ are the images of each other in the straight line l', it follows that the circles $s_{1,2}$ should be the images of each other in the circle l centered at the point O with a radius R with positive power:

$$\mathcal{I}^{+}_{O,R}(s_1) = s_2, \; \mathcal{I}^{+}_{O,R}(s_2) = s_1 \tag{3}$$

or pictorially (Figure 23.2):



We now have three pairs of objects that are the favorable images of each other:

- the pair of circles $s_{1,2}$
- another pair of circles $s'_{1,2}$ and
- a straight line l' and a circle l

That is, if the circles $s_{1,2}$ are the images of each other in the circle $l(O, R)$ with positive power, the circles $s'_{1,2}$ are the images of each other in the straight line l' and the straight line l' is the image of the circle l under an inversion $\mathcal{I}$ with positive power with respect to a circle $c(P \in l, r)$ then the circles $s'_{1,2}$ are the images of the circles $s_{1,2}$ under the same inversion $\mathcal{I}$ correspondingly.

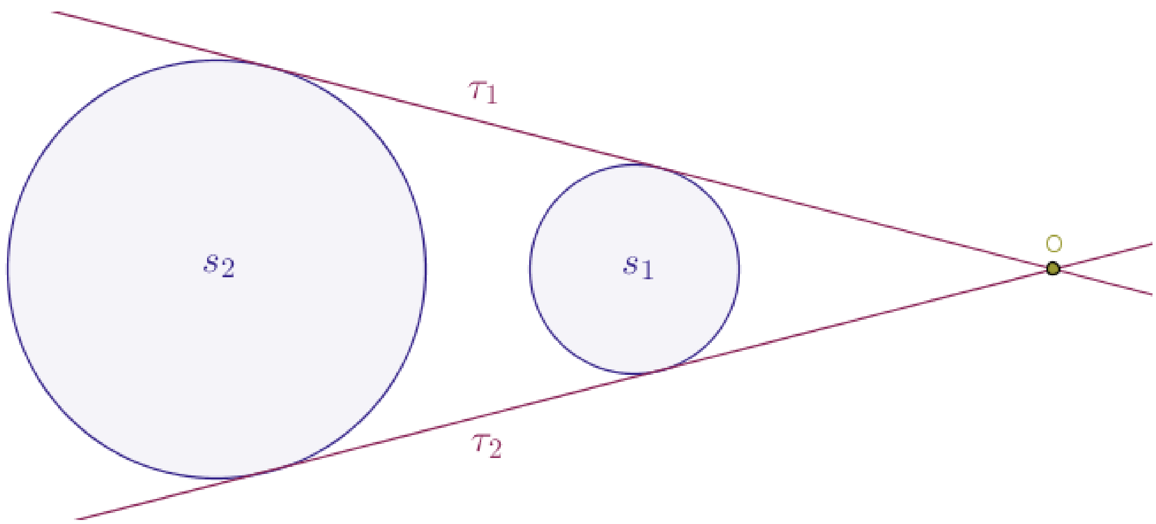
We now have arrived at the original problem statement and we can formulate a forward-order solution sought-after:

- given two coplanar circles $s_{1,2}$ of distinct radii, $r_1 \neq r_2$, not located one inside of the other, find* the center, O, and the positive power, r^2, of inversion of the parent plane under which the circles $s_{1,2}$ are the images of each other
- the said center, O, and power, r^2, of inversion found in the previous step will determine a circle, l, in which the given circles, $s_{1,2}$, are the reflections of each other
- pick any point P such that $P \in l$
- under any inversion of the parent plane $\mathcal{I}^{+}_{P,r}$ the given circles, $s_{1,2}$, with distinct radii will be reflected into two circles, $s'_{1,2}$, with equal radii

Do behold the tactical convenience or, no pun intended, the tactical power of the last observation. Since we are free to choose not only the exact location of the point P on l but the radius, r, of the circle of inversion, c, we can, if need be, choose such a radius r that the circle of inversion c will be orthogonal to the circle:

- s_1, in which case under that inversion $\mathcal{I}^+_{P,r}$ the circle s_1 will be taken into itself and, magically, both inverted circles, $s'_{1,2}$, will have their radii equal to the radius r_1 of the circle s_1
- s_2, in which case under that inversion $\mathcal{I}^+_{P,r}$ the circle s_2 will be taken into itself and, magically, both inverted circles $s'_{1,2}$ will have their radii equal to the radius r_2 of the circle s_2

*In order to find an inversion $\mathcal{I}^+_{O,R}$ under which the given circles are the images of each other, we observe that in this particular case the straight lines $\tau_{1,2}$ that play the roles of common external tangents drawn to the circles $s_{1,2}$ (Figure 23.3):



will always intersect at a point, O, that will play the role of a center of two transformations under which such circles will be the images of each other:

- *homothety* and
- inversion in a circle

The task of constructing the objects $\tau_{1,2}$ is a previously solved, with Euclidean tools alone, problem.

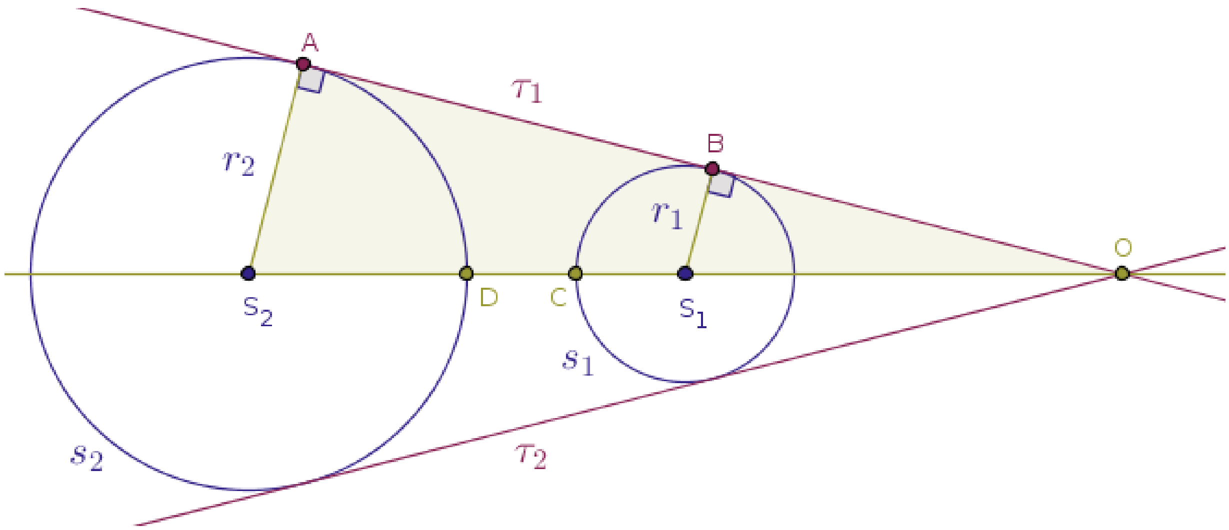
We now prove that the point O thus constructed is the center of an inversion $\mathcal{I}^+_{O,R}$ sought-after.

Let S_1 and S_2 be the centers of the circles s_1 and s_2 correspondingly.

For demonstration purposes, let A and B be the points where the straight line τ_1 touches the circles s_1 and s_2 correspondingly.

Then, evidently the points O, S_1 and S_2 will be collinear, as they will sit on the bisector of the angle formed by the said tangents $\tau_{1,2}$.

Hence, the right, by construction, triangles OS_2A and OS_1B (Figure 23.4):



are similar and in such triangles we have it that:

$$\frac{|OS_2|}{r_2} = \frac{|OS_1|}{r_1}$$

from where:

$$\frac{|OS_2|}{|OS_1|} = \frac{r_2}{r_1} = \text{const} = k > 0$$

for some real number k, implying that:

$$|OS_2| = kr_2, \ |OS_1| = kr_1$$

Let the straight line of centers S_1S_2 intersect the circles $s_{1,2}$ at the points C and D as shown in Figure 23.4 above.

If we demand that the inversion of the parent plane $\mathcal{I}^+_{O,R}$ transforms the circle s_1 into the circle s_2 and vice versa, then the above points C and D must also be the images of each other under $\mathcal{I}^+_{O,R}$.

Constructing the product $|OC| \cdot |OD|$, we see that:

$$|OD| = |OS_2| - r_2 = r_2 \cdot (k - 1)$$

and that:

$$|OC| = |OS_1| + r_1 = r_1 \cdot (k+1)$$

Thus:

$$|OC| \cdot |OD| = r_1 \cdot r_2 \cdot \left(k^2 - 1\right) = \text{const} = R^2 \tag{4}$$

But this situation** is actually much more favorable than its symbolic essence captured in (**4**) may seem to suggest because in Chapter 4, Points, Part 2/2 , in the Problem 7 we already solved exactly the needed problem, and with Euclidean tools alone at that:

given two distinct points, A and A', that are the inverses of each other with respect to a circle with a given center, O, construct the radius of inversion, r, with positive power

Here, the role of the point A above is played by our point, say, C and the role of the point A' above is played by our point D. We also name the radius of inversion sought-after as R.

Altogether, we see that any two coplanar circles $s_{1,2}$ of distinct radii, $r_1 \neq r_2$, not located one inside of the other, can be reflected into two circles of equal radii in two steps:

- in the first step, we find the the center, O, and the radius, R, of the circle, l, under the inversion with respect to which with positive power the given circles $s_{1,2}$ are the images of each other
- in the second step, we make any point P on l the center of a circle, c, under the inversion with respect to which with positive power the given circles $s_{1,2}$ of distinct radii will be reflected into the circles of equal radii

Note how the atomic units of knowledge, the Problem 7 in this case, become the building blocks in the hierarchy of the solutions of more elaborate problems.

**In order to avoid the clutter and not break the flow of our main argument, here we complete the proof of the fact that the external center of homothety O shown in Figure 23.4 doubles up as a center of inversion with positive power under which the input circles $s_{1,2}$ are the images of each other.

We already deduced the relation (**4**) for a pair of “special” points, C and D. We now want to show that that relation will hold of any C-like and D-like points whatsoever. This can be done in a number of different ways. Here is one.

First, we show that the relation (**4**) holds for another pair of special points, A and B. Since $|OB|$ is a tangent to the circle s_1, it follows that for the square of $|OB|$ we have:

$$|OB|^2 = (|OS_1| + r_1) \cdot (|OS_1| - r_1) = |OS_1|^2 - r_1^2 = k^2 r_1^2 - r_1^2 = r_1^2(k^2 - 1)$$

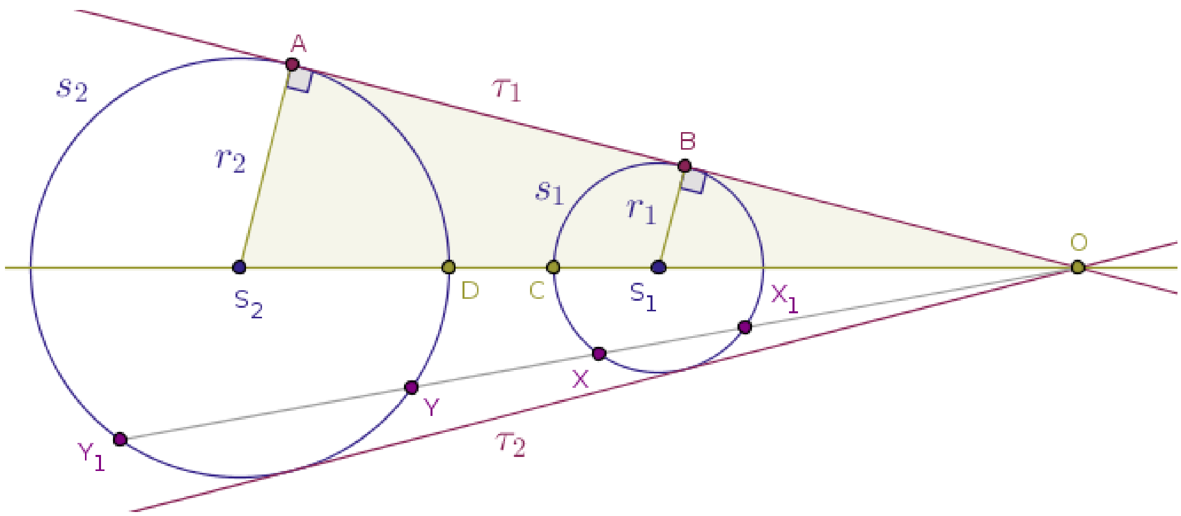
Likewise, since $|OA|$ is a tangent to the circle s_2, for the square of that distance we have:

$$|OA|^2 = (|OS_2| + r_2) \cdot (|OS_2| - r_2) = |OS_2|^2 - r_2^2 = k^2 r_2^2 - r_2^2 = r_2^2(k^2 - 1)$$

and for the product of lengths of these two tangents we have:

$$|OA| \cdot |OB| = \sqrt{r_2^2(k^2 - 1) r_1^2(k^2 - 1)} = r_1 r_2 (k^2 - 1) = |OC| \cdot |OD|$$

Lastly, picking a random point X on the circle s_1 that different from the points C and B (Figure 23.5):



we agree that the straight line OX will intersect the circle s_1 at the point X_1 for the second time and it will intersect the circle s_2 at the distinct points Y and Y_1.

Then, under the corresponding homothety of the parent plane the points X and Y_1 are the images of each other:

$$|OX| \cdot k = |OY_1|$$

and the points X_1 and Y are the images of each other as well:

$$|OX_1| \cdot k = |OY|$$

where k, of course, is the same coefficient of homothety as above, $k = r_2/r_1$.

Constructing the product of distances $|OX| \cdot |OY|$ from above relations, we find:

$$|OX| \cdot |OY| = \frac{|OY_1|}{k} \cdot |OX_1| \cdot k = |OY_1| \cdot |OX_1|$$

Since the straight line OX is a secant for two circles, s_1 and s_2, at once, by the corresponding theorem from the theory of circles, we have for the square of the length of the tangents to these circles we have:

$$|OY_1| = \frac{|OA|^2}{|OY|} \quad \text{and} \quad |OX_1| = \frac{|OB|^2}{|OX|}$$

Thus:

$$|OY_1| \cdot |OX_1| = \frac{|OA|^2 \cdot |OB|^2}{|OY| \cdot |OX|}$$

But we already computed the product of the squares of the lengths of the tangents $|OA|$ and $|OB|$. Hence:

$$|OX| \cdot |OY| = |OY_1| \cdot |OX_1| = \frac{|OA|^2 \cdot |OB|^2}{|OY| \cdot |OX|}$$

from where:

$$|OX| \cdot |OY| = |OA| \cdot |OB| = r_1 r_2 (k^2 - 1) = \text{const} = R^2$$

which precisely means that the points X and Y are the images of each other under the inversion of the parent plane with positive power in a circle that has exactly the point O as its center and the length $\sqrt{r_1 \cdot r_2 \cdot (k^2 - 1)}$ as its radius. □

Two more things.

One. We observe in passing that the input circles $s_{1,2}$ shown in Figure 23.4 also have *an internal* center of homothety, in which case the corresponding coefficient of homothety is *negative* but in that specific case, when the input circles have distinct radii and are not in contact it will not be possible to construct a circle of inversion that is similar to the circle of inversion that has its center at the external center of homothety, O.

If, however, two input circles with distinct radii are located one inside of the other then it is possible to construct the corresponding circle of inversion that will take one of these circles into the other but in this specific case that is possible only for the **in**ternal center of homothety and it is **im**possible to do so for the **ex**ternal center of homothety.

Two. Barring the above two special cases, if we look at our input circles “as a whole” and ignore the tactically immediate individual fates of the points that make up such circles then, via the theorem that we just proved, it becomes evident that inversion of the real plane in a circle and homothety are isomorphic, virtually indistinguishable, essentially or structurally the same.

In other words, from a somewhat less sensitive or from a somewhat numbed down perspective inversion and homothety both take circles into circles.

However and of course, if we treat a circle as a set of points and trace the individual fates of such points then inversion and homothety are different geometric transformations that reflect circles into circles in a different manner.